

Plan of Knowledge: Retrieval-Augmented Large Language Models for Temporal Knowledge Graph Question Answering

Ying Zhang^{1b}, Xinying Qian^{1b}, Yu Zhao^{1b}, Baohang Zhou, Xuhui Sui^{1b}, and Xiaojie Yuan^{1b}

Abstract—Temporal Knowledge Graph Question Answering (TKGQA) aims to answer time-sensitive questions by leveraging factual information from Temporal Knowledge Graphs (TKGs). While previous studies have employed pre-trained TKG embeddings or graph neural networks to inject knowledge, they fail to fully capture the complex semantics of temporal constraints. Recently, Large Language Models (LLMs) have shown remarkable progress, benefiting from their strong semantic understanding and reasoning generalization capabilities. However, their temporal reasoning ability remains limited, often suffering from hallucinations and knowledge gaps. To address these limitations, we propose PoK, a Plan of Knowledge framework equipped with a contrastive temporal retriever. Specifically, the Plan of Knowledge module decomposes a complex temporal question into a sequence of sub-objectives based on predefined operators, providing intermediate guidance for reasoning exploration at the reasoning level. In parallel, we construct a Temporal Knowledge Store (TKS) with a contrastive retrieval framework, enabling the model to retrieve semantically and temporally aligned facts from TKGs, thereby enhancing temporal modeling at the representation level. By combining structured planning with temporal knowledge retrieval, PoK improves the interpretability and factual consistency of temporal reasoning. Extensive experiments on four benchmark TKGQA datasets show that PoK significantly improves retrieval precision and reasoning accuracy, surpassing state-of-the-art methods by up to 35.3 percentage points.

Index Terms—Temporal knowledge graph, knowledge graph question answering, time-sensitive question answering.

I. INTRODUCTION

KNOWLEDGE graphs (KGs) such as DBpedia [1], Freebase [2], and Wikidata [3] organize structured knowledge in the form of triples (*subject, predicate, object*). They have been widely adopted in various downstream applications, including

question answering [4], recommender systems [5], and information retrieval [6]. However, many real-world facts are inherently dynamic and evolve over time. For instance, the triple (*Lionel Messi, member of, FC Barcelona*) became invalid after Messi left FC Barcelona in 2021. To capture such temporal dynamics, temporal knowledge graphs (TKGs) have been introduced, where facts are represented as quadruples (*subject, predicate, object, timestamp*). TKGs such as ICEWS [7] and GDELT [8] continuously record large-scale political, economic, and social events with explicit temporal annotations. These temporal extensions enable time-aware reasoning and support applications in which temporal context plays a crucial role.

Temporal knowledge graph question answering (TKGQA) aims to answer time-sensitive questions by leveraging the knowledge stored in TKGs [9]. Compared with conventional question answering, time-sensitive question answering poses additional challenges, as it requires not only reasoning over entities and relations but also sophisticated temporal inference across evolving events. Traditional approaches [9], [10] cast TKGQA as a temporal KG completion problem, estimating answer likelihoods via scoring functions. Nevertheless, these methods often struggle to fully capture the complex semantic and temporal constraints expressed in natural language questions [11]. In contrast, large language models (LLMs) have demonstrated substantial potential in question answering and information retrieval tasks [12], [13], effectively integrating semantic understanding with knowledge graph reasoning [14], [15]. This motivates the integration of temporal knowledge into LLMs to enhance their capacity for addressing complex, multi-hop temporal questions. However, augmenting LLMs with temporal reasoning capabilities presents several challenges:

1) *Hallucination in multi-hop time-sensitive questions*: Many time-sensitive questions require multi-hop reasoning, such as “After the Danish Ministry, who was the first to visit Iraq?”. Reasoning over such questions is particularly challenging, as LLMs must not only integrate information from multiple hops but also accurately capture temporal constraints such as “first,” “after,” or “before.” As illustrated in Fig. 1, LLMs often hallucinate when dealing with such queries, generating logically inconsistent or factually incorrect reasoning chains. Even when following a step-by-step reasoning paradigm (Fig. 1(1)), hallucinations frequently occur in intermediate steps, leading to erroneous final answers. Moreover, as shown in Fig. 1(4), existing decomposition-based methods express sub-questions in

Received 23 October 2025; revised 2 May 2026; accepted 20 July 2026. Date of publication 14 August 2026; date of current version 15 September 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62272250 and in part by the Natural Science Foundation of Tianjin, China under Grant 22JCJQJC00150. Recommended for acceptance by Z. Wang. (Corresponding author: Xinying Qian.)

Ying Zhang, Xinying Qian, Yu Zhao, Xuhui Sui, and Xiaojie Yuan are with the Tianjin Key Laboratory of Network and Data Security Technology, College of Computer Science, Nankai University, Tianjin 300350, China (e-mail: yingzhang@nankai.edu.cn; qianxinying@dbis.nankai.edu.cn; zhaoyu@dbis.nankai.edu.cn; suixuhui@dbis.nankai.edu.cn; yuanxj@nankai.edu.cn).

Baohang Zhou is with the School of Software, Tiangong University, Tianjin 300380, China (e-mail: zhoubaohang@tiangong.edu.cn).

Digital Object Identifier 10.1109/TKDE.2026.3718295

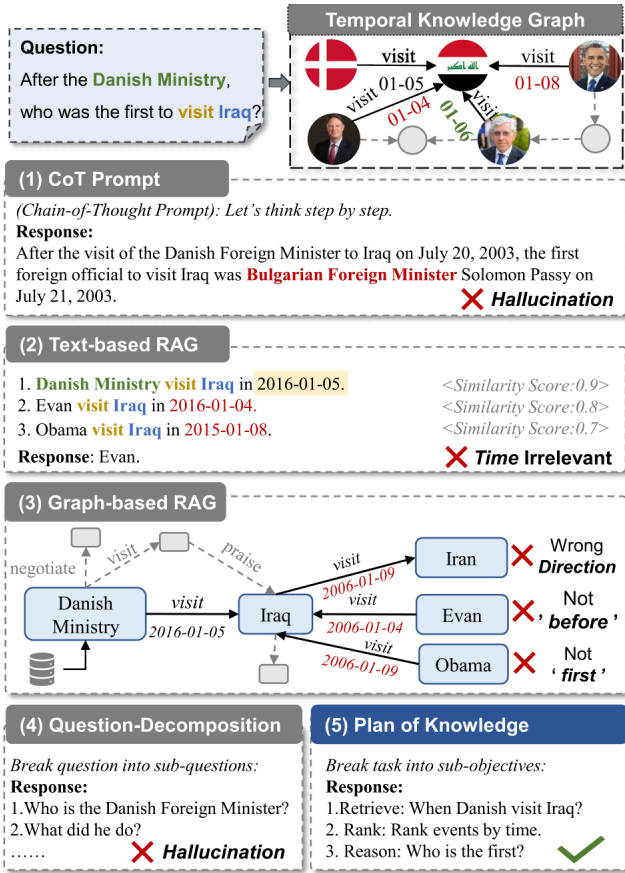


Fig. 1. Illustration of the key differences and challenges in integrating temporal knowledge into large language models across (1) CoT prompting, (2) text-based RAG, (3) graph-based RAG, (4) question-decomposition RAG, and (5) our proposed plan of knowledge RAG framework.

unconstrained natural language, where temporal constraints and semantic reasoning are entangled. Since LLMs struggle with temporal ordering [16], this often leads to temporal inconsistencies and hallucinations. In contrast, as illustrated in Fig. 1(5), explicitly planning a complex temporal question into a sequence of structured sub-tasks effectively decouples these operations. By using predefined executable steps, temporal semantics are preserved at each stage. Therefore, we argue that this explicit *planning of knowledge* is essential for reducing hallucinations and improving the reliability.

2) *Lack of Temporal Knowledge*: Enhancing the temporal reasoning capability of LLMs requires access to relevant knowledge from Temporal Knowledge Graphs (TKGs). Text-based Retrieval-Augmented Generation (RAG) methods [17] often leverage off-the-shelf retrievers such as BGE [18] to extract supporting facts from a knowledge graph as background knowledge. However, as in Fig. 1(2), conventional approaches optimize for semantic similarity while ignoring temporal constraints, leading to semantically plausible but temporally incorrect retrievals. Graph-based RAG methods [12] perform reasoning directly on the graph. Yet, TKGs introduce an additional temporal dimension, making them substantially more complex and often unable to retrieve temporally relevant facts, as in Fig. 1(3). Therefore,

designing a retriever that jointly accounts for both semantic relevance and temporal consistency is crucial for the TKGQA task.

To address these challenges, we propose PoK, a time-aware Plan of Knowledge framework for TKGQA. To mitigate hallucination in multi-hop temporal questions, we introduce an operator-grounded planning strategy that decomposes complex questions into structured sub-objectives over predefined executable operators (*Retrieve*, *Rank*, and *Reason*). This design explicitly decouples temporal ordering from semantic inference at the reasoning level, guiding the stepwise exploration and integration of temporal knowledge from TKGs. However, this reasoning-level decoupling does not eliminate the need for temporal awareness during retrieval. Each retrieval step must still distinguish temporally inconsistent facts from semantically similar but temporally misaligned ones. To this end, we design a time-aware retrieval module that unifies semantic and temporal information at the representation level. Specifically, learnable task prompts are injected into question representations to enhance temporal awareness. Through contrastive fine-tuning with temporally discriminative hard negatives constructed via timestamp and relation corruption, question embeddings are aligned with factual representations across both semantic and temporal dimensions, forming a Temporal Knowledge Store (TKS) for efficient retrieval. During inference, PoK performs retrieval and re-ranking to ensure that selected facts are both contextually relevant and temporally consistent. Finally, the retrieved evidence is fed into a fine-tuned LLM for inference. Experiments on four benchmark datasets demonstrate that PoK substantially improves temporal reasoning, outperforming baselines in both retrieval precision and answer accuracy. Overall, our work makes the following contributions:

- To mitigate hallucination, we integrate LLMs with temporal knowledge and propose PoK, a Plan of Knowledge framework that incorporates predefined temporal operators to guide reasoning over temporal knowledge graphs.
- We design a prompt-based contrastive time-aware retrieval strategy that simultaneously pays attention to semantic similarity and temporal constraints.
- We conduct experiments on four benchmark datasets, demonstrating that our approach significantly improves the retrieval precision and reasoning accuracy, achieving absolute improvements of 1.4, 5.1, 35.3, and 33.7 percentage points compared with state-of-the-art methods.

This article extends our conference version [19] with the following improvements:

- Considering that the previous framework cannot handle multi-hop reasoning, we propose a novel Plan of Knowledge framework.
- We further enhance the temporal retriever with LLM-based semantic representations, task prompt, and an InfoNCE objective, improving both semantic discrimination and temporal sensitivity.
- We expand the related work with a comprehensive review of TKGQA and knowledge-enhanced LLM reasoning, including recent LLM advances, fine-tuning strategies, and RAG paradigms.

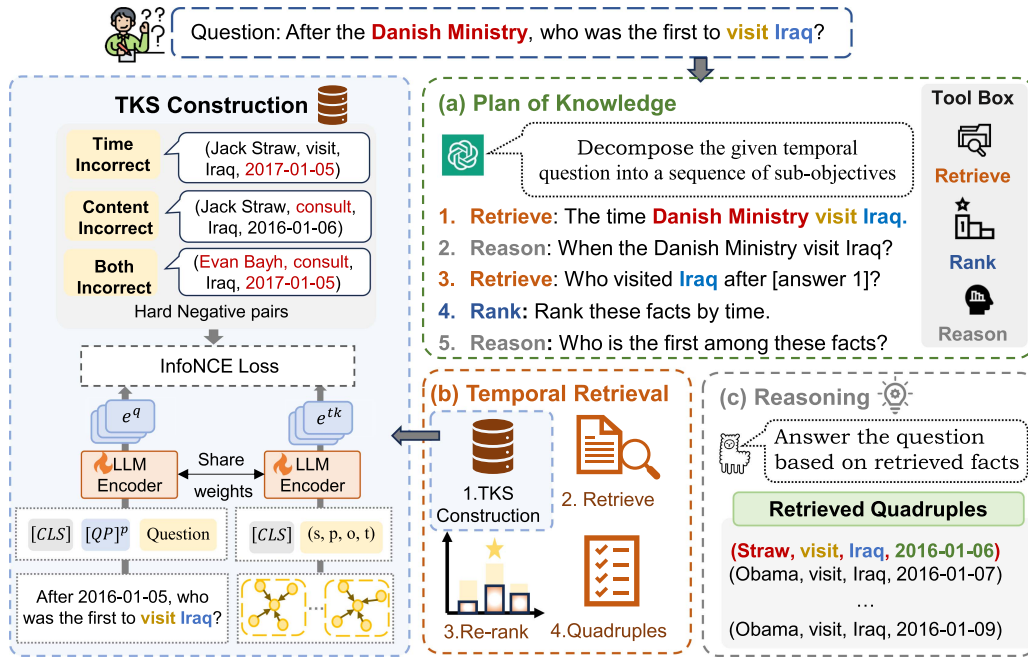


Fig. 2. The overall architecture of PoK consists of three main modules: Plan of Knowledge, Temporal Retrieval, and Reasoning. The left part illustrates the construction of the Temporal Knowledge Store (TKS), which serves as the foundation for retrieval.

- We introduce more datasets and conduct extensive experiments on four datasets, providing deeper analyses of model performance.

The remaining sections of this article are organized as follows. Section II provides a concise review of related work. Section III introduces the problem formulation and notations. Section IV details our proposed framework. Section V presents the experimental setup, main results and additional analysis. Finally, Section VI concludes the paper and discusses potential directions for future work.

II. RELATED WORK

A. Temporal Knowledge Graph Question Answering

Existing Temporal Knowledge Graph Question Answering (TKGQA) methods generally fall into three categories: semantic parsing-based methods, TKG embedding-based methods, and more recently, LLM-based approaches. In this subsection, we focus on the first two task-specific paradigms, while LLM-based methods are discussed in Section II-B.

Semantic parsing-based methods map natural language questions to executable logical forms for precise retrieval. TEQUILA [20] employs rule-based templates to generate SPARQL queries, while SF-TQA [21] captures complex temporal constraints. Although highly interpretable, these methods rely on expensive annotations and exhibit limited generalization to complex temporal expressions.

TKG embedding-based methods formulate TKGQA as a TKG completion task by projecting questions and temporal quadruples into a shared embedding space, and can be broadly categorized into tensor factorization-based and graph neural network (GNN)-based approaches.

Tensor factorization-based methods map TKGs into continuous vector spaces using factorization techniques, such as

Canonical Polyadic (CP) [22] and Tucker decomposition [23]. CronKGQA [9] establishes a basic alignment paradigm by scoring candidate answers via interactions between questions encoded by pretrained language models (PLMs) and TComplEx [24] embeddings. TempoQR [10] extends this idea by jointly integrating contextual, entity, and time-specific features. MultiQA [11] further introduces Transformers to capture multi-granularity temporal dependencies, thereby achieving a more comprehensive modeling of time-sensitive queries.

GNN-based methods focus on structural representation learning through temporal graph propagation. Early approaches like EXAQT [25] and TwiRGCN [26] both utilize graph convolutions to capture multi-hop interactions, with the latter introducing temporal relevance to modulate message passing. Building upon these foundations, LGQA [27] and TMA [28] further enhance reasoning capabilities by adaptively fusing local-global structures and temporal concepts to handle more complex query contexts. M3TQA [29] extends this fusion paradigm into a multi-view and multi-stage framework, providing more robust calibration for heterogeneous graph signals. Most recently, to bridge the gap between structural signals and semantic understanding, GenTKGQA [30] deeply integrates LLMs with GNNs by retrieving and encoding temporal subgraphs as virtual indicators. Despite these advancements, these methods still struggle with complex compositional and numerical temporal reasoning and lack explicit interpretability, prompting a shift toward LLM-centric planning and reasoning.

B. Knowledge-Augmented Large Language Models

Large language models (LLMs) have demonstrated strong generative and reasoning capabilities, which can be further enhanced through prompting strategies such as Chain-of-Thought (CoT) [15]. To extend LLMs to knowledge-intensive tasks,

retrieval-augmented generation (RAG) has emerged as a central paradigm. Existing approaches can be broadly categorized based on how they integrate retrieval with reasoning.

Text-based RAG methods retrieve evidence by semantic similarity matching. Early RAG methods [31] follow a retrieve-then-read pipeline, where dense embeddings are used to calculate the semantic distance between the query and external documents. Subsequent frameworks such as ReAct [32] interleave reasoning with this similarity-based tool use, enabling iterative evidence acquisition for multi-hop questions. However, since these approaches rely strictly on semantic proximity and overlook temporal constraints, their ability to capture time-evolving facts is limited.

Graph-based RAG methods retrieve evidence by traversing the structured topology of KGs. At the retrieval level, G-Retriever [33] formulates this graph extraction as a Prize-Collecting Steiner Tree problem to obtain compact evidence subgraphs. Beyond one-shot retrieval, methods such as Struct-GPT [34] and ToG [35] execute retrieval through progressive graph walking, performing multi-hop reasoning by iteratively exploring interconnected nodes and edges. Despite their strong structural reasoning capabilities, these graph-walking strategies are tailored for static environments. The introduction of the temporal dimension in TKGs increases the complexity of graph traversal, making these methods not directly applicable.

Question Decomposition-based Methods aim to tackle complex multi-hop questions by breaking them down into simpler sub-questions. CoQ [36] utilizes a progressive decomposition approach, while KELDaR [37] constructs a question decomposition tree and performs atomic retrieval at leaf nodes to extract precise KG subgraphs. PoG [38] further introduces self-corrective adaptive planning, enabling LLMs to maintain historical memory and dynamically adjust reasoning paths through reflection. In the temporal domain, RTQA [39] adapts recursive decomposition for TKGs, and ARI [40] prompts LLMs to generate intermediate abstracts prior to knowledge base interaction. However, these methods express sub-questions in unconstrained natural language, where temporal constraints and semantic reasoning are entangled, making it difficult to enforce temporal consistency across steps and leading to hallucinations. To improve reasoning stability, TempAgent [41] incorporates manually designed temporal tools. Nevertheless, introducing a number of customized external operators increases system complexity and demands extensive manual design effort, thereby limiting the flexibility and scalability of the framework.

In contrast, by decoupling temporal ordering from semantic understanding, PoK effectively mitigates hallucinations and enables the retrieval of temporally consistent knowledge, thereby improving robustness in complex temporal reasoning.

III. PRELIMINARIES

Temporal Knowledge Graph (TKG) is defined as $\mathcal{G} = \{\mathcal{E}, \mathcal{P}, \mathcal{T}, \mathcal{F}\}$, where $\mathcal{E}$, $\mathcal{P}$, and $\mathcal{T}$ denote the sets of entities, predicates, and timestamps, respectively. Each temporal fact is a quadruple $(s, p, o, t) \in \mathcal{E} \times \mathcal{P} \times \mathcal{E} \times \mathcal{T}$, with s and o as subject and object, p as the relation, and t as the time of validity. The

TABLE I
PLAN OF KNOWLEDGE PROMPT TEMPLATE

Plan of Knowledge Prompt
Task: Decompose the given question into a sequence of sub-objectives.
Guidelines:
1. Each sub-objective must use one of the predefined operators: Retrieve, Rank, or Reason.
2. Separate each sub-objective with a semicolon (;).
3. Use [answer i] to refer to the output of a previous sub-objective i.
4. Ensure the sub-objectives form a logical reasoning chain.
Example:
Input: Who investigated China first after Segolene Royal?
Output:
Retrieve: When did Segolene Royal investigate China?;
Retrieve: Who investigated China after [answer 1]?;
Rank: Rank these facts by time;
Reason: Who is the first among [answer 2]?;
Now process:
Question: <question>

set of all facts is $\mathcal{F}$. Unlike static KGs, TKGs capture temporal dynamics, enabling reasoning over time-dependent knowledge.

Temporal Knowledge Graph Question Answering (TKGQA) aims to infer answers to time-sensitive questions $q \in \mathcal{Q}$ using temporal facts $f = (s, p, o, t)$ from $\mathcal{G}$.

IV. METHOD

A. Overview

We propose PoK, a time-aware Plan of Knowledge framework for complex temporal reasoning in TKGQA. The core idea is to decompose each temporal question into a structured sequence of sub-objectives grounded in three predefined operators (*Retrieve*, *Rank*, and *Reason*), which explicitly decouples temporal ordering from semantic inference and mitigates hallucination in multi-hop reasoning. To address the lack of temporal knowledge, we further design a time-aware retrieval module that jointly captures semantic relevance and temporal consistency via task-specific prompts and contrastive fine-tuning with hard negatives constructed by corrupting entities, relations, and timestamps. Retrieved facts are stored in a Temporal Knowledge Store (TKS) for efficient access. During inference, a re-ranking module refines candidate facts, and the top-ranked evidence is passed to the *Reason* stage where a fine-tuned LLM generates the final answer. The architecture of PoK is illustrated in Fig. 2.

B. Plan of Knowledge Framework

To systematically support multi-hop and time-sensitive reasoning, we introduce a time-aware *Plan of Knowledge (PoK)* framework that explicitly models the reasoning process as a structured plan. The core idea is to decompose a complex temporal question into a sequence of *sub-objectives*, each corresponding to an interpretable reasoning step.

Formally, given a temporal question q , we prompt the LLM to generate a structured plan of sub-objectives as in (1), which serves as guidance for reasoning. The prompt template used for this decomposition is detailed in Table I.

$$O_q = [(op_i, o_i)]_{i=1}^n = \text{LLM}(\text{Prompt}(q)),$$

$$q \in \mathcal{Q}, \quad op_i \in \{\text{Retrieve}, \text{Rank}, \text{Reason}\}. \quad (1)$$

where O_q denotes the generated sequence of plan objectives and each o_i represents an individual sub-objective. Each sub-objective is grounded in a predefined temporal planning operator op_i , ensuring that the reasoning process remains structured and semantically consistent.

Operator Design: Temporal multi-hop reasoning requires both semantic matching and strict chronological ordering. However, existing methods [39], [40] entangle these two processes in unconstrained natural language sub-questions. Because LLMs struggle with strict numerical constraints and chronological ordering [16], this coupling frequently leads to temporal hallucinations. To address this issue, we decouple the processes into three operators corresponding to the essential stages of temporal multi-hop reasoning: retrieving relevant evidence (*Retrieve*), chronologically ordering candidate facts (*Rank*), and performing logical inference (*Reason*).

This decoupling is particularly important for handling temporal constraints such as *before*, *after*, and *first*, which require explicit chronological ordering. When such ordering is handled implicitly by LLMs, it is prone to inconsistencies. Therefore, we implement *Rank* as a deterministic symbolic function, avoiding sensitivity to numerical reasoning and ensuring reliable temporal ordering.

- *Retrieve* (Section IV-C): A mapping $f_{\text{Retrieve}} : \mathcal{O} \rightarrow 2^{\mathcal{F}}$, which retrieves a subset of relevant temporal facts $\mathcal{F}$ for a given sub-objective $o_i \in \mathcal{O}$. Here, $2^{\mathcal{F}}$ denotes the power set of $\mathcal{F}$.
- *Rank*: A temporal sorting function $f_{\text{Rank}} : 2^{\mathcal{F}} \rightarrow \mathcal{F}^*$, which orders the retrieved facts chronologically (in either ascending or descending order), producing an ordered sequence $\mathcal{F}^*$.
- *Reason* (Section IV-D): A compositional inference function $f_{\text{Reason}} : \mathcal{O} \times \mathcal{F}^* \rightarrow \mathcal{A}$, which integrates temporally ordered evidence guided by the sub-objective o_i to infer the final answer $a \in \mathcal{A}$.

By decoupling temporal ordering from semantic inference, PoK enables more stable and interpretable reasoning. The framework grounds each sub-objective in a predefined set of executable operators, ensuring that every reasoning step preserves the temporal semantics of the original query, thereby yielding a more controllable and transparent reasoning process.

C. Temporal Retrieval Module

With the Plan of Knowledge decomposing each question into structured sub-objectives, the *Retrieve* operator serves as the first executable step, responsible for grounding each sub-objective in relevant factual evidence from the TKG. However, on some TKGs such as ICEWS [42], entity-linking tools are unavailable [11], making the conventional pipeline paradigm [11], [27], which relies on entity-linking tools to identify entities and relations before retrieving candidate facts, infeasible. We therefore adopt the direct retrieval paradigm [43], which matches questions directly with factual embeddings.

Building on this idea, we propose a prompt-based contrastive temporal retrieval framework that captures both semantic similarity and temporal constraints. The temporal retrieval module first applies prompt-guided temporal encoding to inject task prompts into question representations, capturing temporal constraints. It then performs contrastive time-aware fine-tuning to align semantically and temporally enriched question embeddings with factual embeddings. Temporal facts are encoded into the Temporal Knowledge Store (TKS) for efficient access, followed by retrieval and re-ranking to obtain the final candidate facts.

Prompt-guided Temporal Encoding: To enhance the encoder's sensitivity to temporal constraints, we integrate task soft prompts inspired by prompt learning [44]. These learnable prompt tokens are inserted into the input sequence and guide the model's attention toward temporal constraints. Specifically, the prompts are represented as a sequence of learnable vectors: $\mathbf{P} = [\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_m]$, $\mathbf{p}_i \in \mathbb{R}^d$, where m denotes the number of prompt tokens. Given a question $q = [w_1, w_2, \dots, w_n]$, the input sequence is formed as: $\mathbf{X} = [\mathbf{p}_q, \mathbf{e}_{w_1}, \dots, \mathbf{e}_{w_n}]$, and the encoded question embedding is obtained as:

$$\mathbf{E}_{q_i} = LM_t(\mathbf{X}) \in \mathbb{R}^d. \quad (2)$$

Contrastive Time-aware Fine-tuning: Temporal question answering requires candidate facts to satisfy both semantic relevance and temporal constraints, which semantic-only retrievers cannot capture. To address this, we introduce a contrastive learning objective that jointly optimizes semantic and temporal alignment in a shared embedding space.

For each positive question–fact pair, we construct three types of negative samples to model temporal and semantic discrepancies: time-incorrect facts by replacing timestamps, content-incorrect variants by modifying relations, and both-incorrect samples by altering entities and timestamps simultaneously. This strategy encourages the model to capture fine-grained temporal cues and distinguish facts that are semantically plausible but temporally invalid. Following [45], we adopt the InfoNCE loss to jointly model semantic relevance and temporal consistency. Given a question embedding q , a positive fact embedding f^+ , and a set of negative fact embeddings $\mathcal{N}$, the loss is defined as:

$$\mathcal{L} = -\log \frac{\exp(\text{sim}(q, f^+)/\tau)}{\exp(\text{sim}(q, f^+)/\tau) + \sum_{f^- \in \mathcal{N}} \exp(\text{sim}(q, f^-)/\tau)}, \quad (3)$$

where $\text{sim}(\cdot, \cdot)$ denotes cosine similarity, τ is a temperature parameter, and $\mathcal{N}$ is the set of negative samples constructed by corrupting entities, relations, or timestamps.

Temporal Knowledge Store Construction: To support efficient retrieval, we construct a Temporal Knowledge Store (TKS) that encodes both semantic and temporal information for facts in the TKGs. Specifically, each quadruple (s, p, o, t) is converted into a textual template according to its temporal type. For time-point quadruples, the template is "{subject} {predicate} {object} at {time_point}", and for time-interval quadruple, it is "{subject} {predicate} {object}

from $\{\text{begin_time}\}$ to $\{\text{end_time}\}$ ". After encoding, all temporal facts $T(s, p, o, t) \in \mathcal{G}$ are stored as:

$$\text{TKS} = \{\mathbf{E}_f \mid \mathbf{E}_f = LM_t(T(s, p, o, t)), (s, p, o, t) \in \mathcal{G}\}. \quad (4)$$

This dense repository acts as a neural memory that can be efficiently queried through vector similarity search. By representing all facts in a unified embedding space, TKS provides a continuous interface between queries and structured temporal knowledge, enabling fast retrieval during inference.

Temporal retrieval and reranking: During inference, relevant temporal facts are retrieved by computing the cosine similarity between the encoded question $\mathbf{E}_q$ and all fact embeddings $\mathbf{E}_f$ stored in TKS:

$$\phi_{TKS} = \cos(\mathbf{E}_{q_i}, \mathbf{E}_f). \quad (5)$$

For scalability, we employ FAISS [46] for efficient vector search and indexing.

To improve temporal relevance, a time-filtering function is applied for questions. For each quadruple (s, p, o, t) in the TKG $\mathcal{G}$, we compute the time difference from the query time t_q , filter out quadruples outside the valid range, and normalize the differences to produce the filtered results. Equation (6) illustrates the function for "before"-type questions.

$$\phi_t(t_q, t) = \begin{cases} 1 - \frac{|t_q - t|}{\max(t_q - t)}, & \text{if } (t_q - t) > 0, \\ -\infty, & \text{otherwise.} \end{cases} \quad (6)$$

The final retrieval score jointly integrates semantic and temporal signals:

$$\phi(q, t) = \mu \cdot \phi_{TKS}(\mathbf{E}_{q_i}, \mathbf{E}_t) + (1 - \mu) \cdot \phi_t(t_q, t), \quad (7)$$

$$\mathbf{f} = \arg \max \phi(q_i, t), \quad (8)$$

where μ is a balancing coefficient controlling the trade-off between semantic relevance and temporal coherence. This integrated scoring mechanism ensures that the retrieved evidence is not only linguistically plausible but also temporally valid.

D. Reasoning

With the Plan of Knowledge guiding the decomposition and the Temporal Retrieval Module supplying temporally grounded evidence for each sub-objective, the *Reason* operator serves as the final stage, integrating the retrieved facts to produce the answer. We cast the reasoning process as an optimization problem over the LLM, where the goal is to maximize the likelihood of producing the correct answer a given the question q and evidence. Formally, the objective is defined as:

$$\mathcal{L} = \max_{\Phi} \sum_{(q_i, a) \in \hat{\mathcal{Q}}} \sum_{t=1}^{|a|} \log P_{\Phi}(a_t \mid (q_i, f^+), a_{<t}), \quad (9)$$

where Φ denotes the parameters of the LLM, (q_i, f^+) represents the question q paired with its most relevant fact, and $a_{<t}$ is the partial sequence generated up to step $t-1$. This formulation allows the model to jointly reason over both the context of the question and the factual knowledge retrieved from the TKS. To

TABLE II
REASONING PROMPT TEMPLATE

Reasoning Prompt
Based on the facts, please answer the given question. Keep the answer as simple as possible and return the answers as a list. Facts: <facts> Question: <question>

TABLE III
DATA SPLITS FOR THE DATASETS

Dataset	Type	Train	Val	Test
MULTITQ	Single	283,482	41,735	38,864
	Multiple	103,305	16,244	15,720
	Total	386,787	57,979	54,584
TimeQuestions	Explicit	3,904	1,302	1,311
	Implicit	871	291	292
	Temporal	3,222	1,073	1,067
	Ordinal	1,711	570	567
	Total	9,708	3,236	3,237
Timeline-ICEWS	Simple	17,982	5,994	5,994
	Medium	15,990	5,330	5,330
	Complex	19,652	6,550	6,550
	Total	53,624	17,874	17,874
Timeline-CronQuestion	Simple	7,200	2,400	2,400
	Medium	8,252	2,751	2,751
	Complex	9,580	3,193	3,193
	Total	25,032	8,344	8,344

guide the LLM in generating final answers, we design a simple instruction prompt in Table II.

V. EXPERIMENTS

In this section, we first describe the experimental setup, including the datasets, evaluation metrics, and baseline methods. Then, we conduct extensive experiments on four real-world datasets to answer the following research questions (RQs):

- RQ1: Does PoK outperform existing baselines on TKGQA tasks, including complex multi-hop questions?
- RQ2: What is the contribution of each component to the overall performance of PoK?
- RQ3: How robust and efficient is PoK under varying backbones, retrieval strategies, and hyperparameter settings?
- RQ4: What insights can be drawn from case studies and error analysis?

A. Experiment Settings

1) *Datasets:* To evaluate the performance of PoK, we conduct experiments on four popular datasets, MULTITQ [11], TimeQuestions [25], Timeline-ICEWS, and Timeline-CronQuestions [47]. The statistical information is presented in Table III.

MULTITQ is the largest publicly available TKGQA dataset, constructed from ICEWS05-15 [7], and contains more than 500 K unique question-answer pairs. A distinctive feature of MULTITQ is that it covers multiple temporal granularities,

including year, month, and day, with time spans ranging from 2005-01-01 to 2015-12-31.

TimeQuestions is another widely used and challenging TKGQA benchmark, constructed from temporal facts in Wikidata. Each fact is represented in the form of {subject, predicate, object, begin_time, end_time}, thereby providing rich temporal information for reasoning. Compared with MULTITQ, *TimeQuestions* is smaller in size and only supports year-level temporal granularity. Nevertheless, it spans a much broader temporal range, covering facts across more than 1,600 years.

Timeline-ICEWS and *Timeline-CronQuestions* are datasets constructed from ICEWS Coded Event Data (Time Range) and CronQuestion knowledge graph (Time Point) using TimelineKGQA categorization framework.

2) *Parameter Setting*: For the **planning framework**, we leverage the OpenAI API¹ with the gpt-4o² model to decompose questions into sub-objectives that guide temporal reasoning. For the **temporal retriever**, we fine-tune Qwen3-Embedding-0.7B [48] for 2 epochs, which offers a lightweight yet highly effective solution for semantic retrieval. Each training question is optimized with in-batch negatives as well as three hard negatives. We set the temperature $\tau = 0.01$. During the re-ranking stage, we introduce a balancing coefficient $\mu = 0.2$ to combine semantic relevance with temporal consistency. For the **reasoning backbone**, we conduct a comparative study among several LLMs, including GPT-4o, Qwen3-8B, and LLaMA2-Chat-7B [49]. Based on both performance and efficiency considerations, we ultimately adopt LLaMA2-Chat-7B as our reasoning backbone. The model is fine-tuned for 2 epochs on 2 NVIDIA A6000 GPUs. Due to the considerable size of the MULTITQ dataset, we randomly sample 20% of the training data for fine-tuning. During inference, we retrieve the 20 candidate facts from the retriever and feed them into the LLM. We use Hits@k as the evaluation metric. Hits@k refers to the percentage of correct answers ranked in the top k predictions. Higher Hits@k demonstrates better performance.

3) *Baselines*: To validate the effectiveness of the proposed model, we compare the proposed PoK with three groups of baselines on MULTITQ:

- *Pre-trained LM-based methods*: This category includes BERT [50], DistilBERT, and ALBERT [51]. For BERT and its variants, we generate question embeddings, concatenate them with entity and temporal embeddings, and apply a projection layer, following prior work [11].
- *TKGQA methods*: This category includes EmbedKGQA [52], CronKGQA [9], and MultiQA [11].
- *LLM-based methods*: This category includes LLaMA2-7B, ChatGPT,³ ARI [40], Naive RAG [31], ReAct RAG Agent [32], TempAgent [41], TimeR⁴ [19], and RTQA [39]. For LLaMA2 and ChatGPT, we directly feed the questions as input without additional explanations.

Three groups of baselines are used for comparison on TimeQuestions:

¹ <https://platform.openai.com/docs/api-reference>

² <https://platform.openai.com/docs/models/gpt-4o>

³ <https://platform.openai.com/docs/models/gpt-3-5-turbo>

TABLE IV
HITS@1 PERFORMANCE COMPARISON OF DIFFERENT MODELS ON MULTITQ (%)

Model	Overall	Question Type		Answer Type	
		Single	Multiple	Entity	Time
BERT	8.3	9.2	6.1	10.1	4.0
DistilBERT	8.3	8.7	7.4	10.2	3.7
ALBERT	10.8	11.6	8.6	13.9	3.2
EmbedKGQA	20.6	23.5	13.4	29.0	0.1
CronKGQA	27.9	33.7	13.4	32.8	15.6
MultiQA	29.3	34.7	15.9	34.9	15.7
LLaMA2-7B	18.5	22.0	10.1	23.9	5.5
ChatGPT	10.2	14.7	7.7	13.7	2.0
ARI	38.0	68.0	21.0	39.4	34.4
Naive RAG	37.9	46.9	15.5	24.2	67.2
ReAct RAG	39.8	50.6	13.0	24.3	73.5
TempAgent	70.2	85.7	31.6	62.4	87.0
TimeR ⁴	72.8	88.7	33.5	63.9	94.5
RTQA	<u>76.5</u>	<u>90.2</u>	42.4	<u>69.2</u>	<u>94.2</u>
PoK	77.9	92.9	<u>40.9</u>	69.6	96.2

- *Static KGQA methods*: including PullNet [4], Uniqorn [53], and GRAFT-Net [54].
- *TKGQA methods*: including CronKGQA [9], TempoQR [10], EXAQT [25], LGQA [27], TwiRGCN [26], TMA [28], and M³TQA [29].
- *LLM-based methods*: This category includes LLaMA2-7B, ChatGPT, GenTKGQA [30], and TimeR⁴ [19]. For LLaMA2 and ChatGPT, we also directly feed the questions as input without additional explanations.

For Timeline-ICEWS and Timeline-CronQuestions, we compare our model with RTQA and RAG baselines. The RAG baseline encodes both the knowledge and queries using OpenAI's text-embedding-3-small model, followed by a semantic similarity search to retrieve the most relevant fact.

B. Main Results (RQ1)

We evaluate PoK against all baselines across the benchmark datasets. We first report overall performance on standard temporal QA benchmarks, then examine how PoK handles complex multi-hop temporal questions that require reasoning over multiple temporal constraints.

1) *Overall Performance*: We present the experimental results in comparison with other methods on the MULTITQ, TimeQuestions, Timeline-CronQuestions, and Timeline-ICEWS datasets, as shown in Tables IV, V, and VI. In these tables, the best results are highlighted in bold, while the second-best results are underlined. Across all experimental settings, PoK consistently achieves the best performance, clearly demonstrating its effectiveness and robustness.

For the MULTITQ dataset, PoK achieves state-of-the-art performance. Specifically, we find that PLMs (BERT, ALBERT) and LLMs (LLaMA2, ChatGPT) exhibit the lowest performance on the TKGQA task. This might be due to the lack of necessary temporal knowledge, thus leading to errors in reasoning. Compared with the best-performing traditional KGQA method, PoK

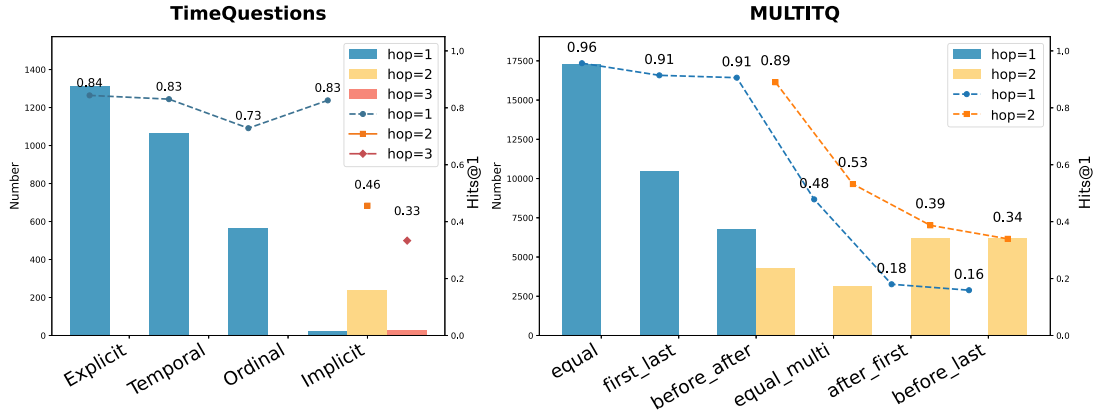


Fig. 3. Comparison of Question Hops and Hit@1 Results across Different Question Types under the PoK Framework.

TABLE V
HITS@1 PERFORMANCE COMPARISON OF DIFFERENT MODELS ON
TIMEQUESTIONS (%)

Model	Overall	Question Type		Answer Type	
		Explicit	Implicit	Ordinal	Temporal
PullNet	10.5	2.2	8.1	2.9	23.4
Uniqorn	33.1	31.8	31.6	20.2	39.2
GRAFT-Net	45.2	44.5	42.8	32.2	51.5
CronKGQA	46.2	46.6	44.5	36.9	51.1
TempoQR	41.6	46.5	3.6	34.9	40.0
EXAQT	57.2	56.8	51.2	42.0	64.2
TwirGCN	60.5	60.2	58.6	51.8	64.1
LGQA	52.9	53.2	50.6	40.2	60.5
TMA	43.5	44.2	41.9	35.2	47.6
M^3 TQA	53.6	53.6	51.0	40.8	61.2
LLaMA2-7B	27.1	26.8	32.5	23.4	27.9
ChatGPT	45.9	43.3	51.1	48.1	46.5
GenTKGQA	58.4	59.6	61.1	57.8	56.3
TimeR ⁴	78.1	82.3	73.0	64.9	83.0
PoK	83.2	85.8	85.3	74.3	84.2

achieves an absolute Hits@1 improvement of 48.6 percentage points, highlighting its capability to handle temporally complex queries. Moreover, compared with recent LLM-based methods, PoK achieves absolute Hits@1 improvements of 39.9 and 1.4 percentage points over ARI and RTQA, respectively. These results validate the strength of our proposed PoK framework in temporal retrieval and reasoning. In addition, PoK also surpasses retrieval-augmented methods such as Naive RAG and ReAct RAG, which further highlights the effectiveness of our temporal retriever.

On the TimeQuestions dataset, PoK achieves the best results across all question categories. KGQA-based methods perform the worst, as they lack the ability to retrieve or reason over temporal facts. LLMs such as GPT-4o and LLaMA2 perform better on TimeQuestions than on MULTITQ, likely because the dataset is constructed on Wikidata [3], which is heavily represented in their pre-training corpora and provides partial knowledge. Moreover, PoK surpasses GenTKGQA, further demonstrating the reliability and generalizability of its temporal retriever and Plan of Knowledge framework.

On Timeline-CronQuestions and Timeline-ICEWS, PoK achieves absolute overall gains of 35.3 and 33.7 percentage points over the strongest baselines, respectively. Notably, for complex questions, our method achieves substantial absolute gains of 30.7 and 22.5 percentage points, further highlighting its effectiveness and superiority.

2) *Performance on Complex Temporal Questions:* We analyze reasoning hops and Hits@1 across question types on MULTITQ and TimeQuestions, as shown in Fig. 3. We do not report additional results on Timeline-ICEWS and Timeline-CronQuestions, as these datasets explicitly define complexity by hop count (1-hop, 2-hop, and 3-hop corresponding to simple, medium, and complex questions). In contrast, MULTITQ and TimeQuestions do not provide such annotations.

The results show that multi-hop reasoning is mainly required for harder cases, such as *Implicit* questions in TimeQuestions and *equal multi*, *before last*, and *after first* in MULTITQ, while simpler questions are typically resolved in a single hop. This suggests that our Plan of Knowledge framework can effectively determine when multi-step reasoning is necessary.

In terms of Hits@1, performance consistently decreases as the number of hops increases. Single-hop questions achieve the highest accuracy, whereas two- and three-hop questions exhibit a clear decline, reflecting the increasing difficulty of multi-hop temporal reasoning. Nevertheless, the model maintains competitive performance on complex cases, indicating that the PoK decomposition effectively supports multi-step reasoning.

C. Ablation Study (RQ2)

In this section, we conduct ablation studies to evaluate the effectiveness of the proposed model. The results are reported in Table VII.

Effect of the PoK-plan Operator: To examine the contribution of the planning component, we remove PoK-Plan and instead directly retrieve facts using the original questions. On multiple-hop questions specifically, the drop is substantially larger at 18.7 percentage points (from 40.9% to 22.2%), demonstrating that structured planning is indispensable for complex temporal reasoning. The drop in TimeQuestions is less significant, mainly

TABLE VI
PERFORMANCE COMPARISON OF DIFFERENT MODELS (IN PERCENTAGE) ON TIMELINEKGQA

Model	CronQuestions KG				ICEWS Actor			
	Overall	Simple	Medium	Complex	Overall	Simple	Medium	Complex
RAG baseline	23.5	70.4	9.2	0.9	26.5	66.0	12.8	1.1
LLaMA2-7B	16.9	4.9	14.3	28.2	11.1	3.5	6.6	32.2
GPT-4o	20.6	6.9	13.0	37.6	11.3	5.1	3.5	35.3
RTQA	29.8	60.8	21.8	13.5	-	-	-	-
PoK	65.1	73.7	53.9	68.3	60.2	74.4	45.6	57.8

TABLE VII
RESULTS OF THE ABLATION STUDY. "W/O" MEANS REMOVING THE MODULE.

Model	MULTITQ					TimeQuestions				
	Overall	Single	Multiple	Entity	Time	Overall	Explicit	Implicit	Ordinal	Temporal
PoK	77.9	92.9	40.9	69.6	96.2	83.2	85.8	85.3	74.3	84.2
w/o PoK-plan	71.3	91.1	22.2	62.6	92.5	82.4	85.7	79.1	73.7	84.0
w/o PoK-rank	75.5	92.3	34.1	67.3	95.5	81.1	84.4	70.6	71.8	83.7
w/o PoK-retrieve	32.0	36.3	21.2	39.7	13.1	44.7	44.4	55.5	45.0	42.0
w/o task prompt	74.9	90.0	37.8	68.3	91.1	82.1	85.0	77.4	74.1	83.4
w/o re-rank	71.0	87.4	30.5	62.2	92.6	79.7	84.4	57.3	72.8	83.0

because its questions concern events that are topically sparse, making it straightforward to retrieve all relevant facts for a given entity without multi-step planning. In contrast, MULTITQ covers dense political events where entities are involved in many temporally overlapping facts, making planning essential for identifying the correct answer.

Effect of the PoK-rank Operator: We further analyze the rank operator in PoK's reasoning module. Specifically, during inference, we randomly shuffle the facts instead of preserving their chronological order. The results demonstrate a clear performance degradation after shuffling, particularly for ordinal questions, indicating that maintaining the temporal order of facts is essential for enabling LLMs to effectively comprehend and utilize temporal knowledge.

Effect of the PoK-retrieve Operator: To further evaluate the role of the PoK-retrieve Operator, we remove the retrieval component and let the model rely solely on the original questions without accessing temporally relevant facts. The results show a substantial drop in performance, confirming that temporal retrieval plays a critical role. By explicitly retrieving relevant temporal facts, the strategy helps LLMs ground their answers, thereby mitigating hallucinations and improving accuracy on implicit temporal questions.

Effect of the task prompt module:

Effect of re-ranking strategy: Removing the re-ranking strategy resulted in a significant decrease in model performance, indicating that filtering out irrelevant information is crucial.

D. In-Depth Analysis (RQ3)

We conduct a series of experiments to further investigate the factors that influence the effectiveness of PoK. Specifically, we examine its generalizability across different LLM backbones,

the impact of alternative retrieval strategies, its sensitivity to the number of retrieved facts, and its runtime efficiency.

1) *Generalizability Across LLMs:* We compare PoK with other LLMs (LLaMA2-7B, GPT-4o, ChatGPT, and Qwen3) on both datasets, as shown in Table VIII. Here, LLMs w/ PoK denote backbone models equipped with PoK strategy and the planned questions.

When using LLMs alone, their performance is much higher on TimeQuestions than on MULTITQ. This is because TimeQuestions largely draws from Wikidata, which is part of many LLMs' pretraining corpora, allowing them to answer directly from prior knowledge. In contrast, MULTITQ involves more specialized political and event-centric temporal knowledge that is rarely covered in pretraining, making it more challenging.

With our PoK strategy, LLMs w/ PoK achieve substantially better results, showing that while LLMs have some inherent temporal reasoning ability, their precision and robustness are limited. PoK explicitly structures retrieval and reasoning, overcoming these limitations and consistently improving performance across backbones, demonstrating its generality. The relatively poor performance of *GPT-4o w/ PoK* is possibly due to a mismatch between the output data format and the ground truth. We discuss this issue in Section V-E. We also observe that fine-tuned LLaMA2 nearly doubles the performance of the untuned model, indicating that fine-tuning effectively constrains the output space and strengthens temporal reasoning.

2) *Effectiveness of the Retrieval:* To thoroughly evaluate the effectiveness of our temporal retrieval approach, we conduct a comparative analysis across several representative models, including Sentence-BERT, fine-tuned Sentence-BERT, Qwen-0.7B, Qwen-4B, and our proposed PoK temporal retriever. Here, the fine-tuned Sentence-BERT is trained using the same data, strategies, and objectives as PoK temporal retrieval to ensure a

TABLE VIII
 EFFECTS OF INTEGRATING THE PoK FRAMEWORK WITH DIFFERENT LLMs FOR REASONING.

Model	MULTITQ					TimeQuestions				
	Overall	Single	Multiple	Entity	Time	Overall	Explicit	Implicit	Ordinal	Temporal
PoK	77.9	92.9	40.9	69.6	96.2	83.2	85.8	85.3	74.3	84.2
LLaMA2-7B	18.5	22.0	10.1	23.9	5.5	27.1	26.8	32.5	23.4	27.9
LLaMA2 w/ <i>finetuned</i>	33.9	22.7	38.4	45.0	7.8	45.8	44.4	46.0	51.9	37.8
ChatGPT	10.2	14.7	7.7	13.7	2.0	45.9	43.3	51.1	48.1	46.5
LLaMA2-7B w/ <i>PoK</i>	58.4	65.6	39.5	60.9	52.3	61.3	62.5	53.4	36.2	75.4
Qwen3-8B w/ <i>PoK</i>	63.1	73.9	36.6	66.4	55.2	64.8	67.5	53.1	39.5	77.3
GPT-4o w/ <i>PoK</i>	<u>66.0</u>	<u>76.7</u>	<u>39.6</u>	<u>67.5</u>	<u>62.3</u>	<u>67.1</u>	<u>70.2</u>	<u>57.5</u>	<u>45.5</u>	<u>77.3</u>

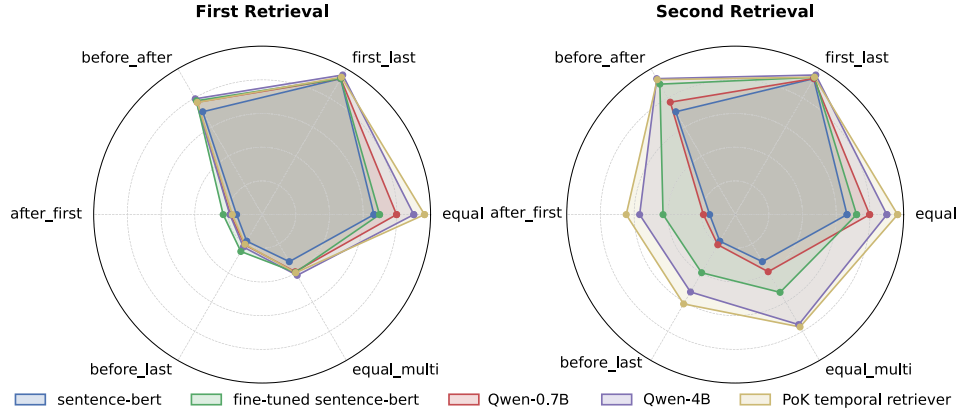


Fig. 4. Comparison of Retriever Performance on the MULTITQ Dataset, including Sentence-BERT, Fine-tuned Sentence-BERT, Qwen-0.7B, Qwen-4B, and PoK Temporal Retriever.

fair comparison. We assess their answer coverage on the MULTITQ dataset and visualize the results for different categories of temporal questions in Fig. 4.

The categories *first_last*, *before_after*, and *equal* represent relatively simple temporal reasoning problems. In these cases, the performance gap among different models is relatively small; however, our method still surpasses other retrievers and achieves strong results even in the first retrieval step.

In contrast, the situation changes in the second retrieval stage, which involves more complex and challenging categories such as *after_first*, *before_last*, and *equal_multi*. Here, the overall performance improves substantially compared with the first retrieval, but more importantly, the differences between models become much more pronounced. Notably, our fine-tuned PoK temporal retriever enables the Qwen-0.7B model to outperform the much larger Qwen-4B model, underscoring the critical role of contrastive, temporally-aware fine-tuning.

In contrast, models without fine-tuning (e.g., Sentence-BERT and Qwen-0.7B) exhibit almost no improvement across retrieval rounds. This observation suggests that, even under the Plan of Knowledge framework, smaller embedding models lack the capacity to effectively capture and reason over fine-grained temporal constraints. By integrating contrastive fine-tuning, our approach explicitly encourages the embedding model to learn temporal distinctions and attend more carefully to time-sensitive relational cues. These results collectively demonstrate that time-aware fine-tuning is essential for enabling retrieval models to achieve robust and accurate temporal reasoning.

 TABLE IX
 COMPARISON OF API CALL COUNTS, RUNTIME, AND PROMPT LENGTH ACROSS METHODS

Method	Avg Time (s)	API Calls	Prompt Length
RTQA	3.58	3.96	1185 + 13.01 + 50 × 67.2
PoK	2.18	1	134 + 13.01 + 20 × 67.2

3) *Number of Retrieved Facts*: To examine how the number of retrieved facts affects performance, we vary n and report performance and answer coverage in Fig. 5. It is evident that the model achieves its peak performance when provided with 20 relevant facts. Notably, performance drops slightly at $n = 25$ despite higher coverage, implying that excessive facts introduce noise, whereas too few facts limit context. Thus, $n = 20$ strikes the best balance.

4) *Runtime Efficiency*: To evaluate efficiency, we compare PoK with the recent baseline RTQA on MULTITQ in terms of average runtime, API calls, and prompt length (Table IX). PoK's total runtime consists of training (0.65 s), retrieval (0.06 s), planning (0.95 s), and reasoning (0.52 s). By fine-tuning only a subset of the training data, PoK not only reduces computational cost but also achieves efficient training. Moreover, inference after fine-tuning is faster than that of the base model. Each question in PoK requires just one API call, used solely in the Plan of Knowledge stage; retrieval and reasoning are fully handled by our fine-tuned open-source model.

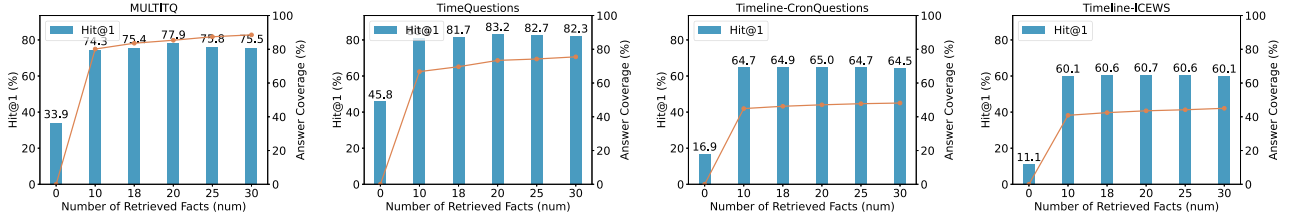


Fig. 5. The Hits@1 results of different retrieved fact numbers across four datasets.

TABLE X
EXAMPLE OF A 3-HOP REASONING PROCESS IN PoK AND GPT-4o

Type	Overlap, 3-hops
Question	Who held a position in the 4th United States Congress and was Secretary of State during Andrew Jackson's presidency?
Answer	Martin Van Buren, Edward Livingston, Louis McLane.
Plan	(1) Retrieve & Reason: [time] = When is Andrew Jackson's presidency? (2) Retrieve & Reason: [Person] = Who was the Secretary of State during [time]? (3) Retrieve & Reason: Who held a position in the 4th United States Congress among [person]?
PoK Response	(1) [1829,1837] (2) ['Martin Van Buren', 'Edward Livingston', 'Louis McLane', 'John Forsyth'] (3) ['Martin Van Buren', 'Edward Livingston', 'Louis McLane']
GPT-4o Response	- John Forsyth - Martin Van Buren

Prompt length further contributes to efficiency. PoK uses a fixed template (134 tokens), the question itself (13.01 tokens on average), and 20 retrieved facts (67.2 tokens each). In contrast, RTQA constructs custom prompts for each question type and retrieves up to 50 facts, leading to much longer inputs.

E. Case Study and Error Analysis (RQ4)

1) *Case Study*: We illustrate PoK's and GPT-4o's reasoning behaviors on overlap-type 3-hop and after/first-type 2-hop questions in Tables X and XI, respectively. From the cases, we observe that for complex temporal questions, the Plan of Knowledge framework enables step-by-step reasoning rather than attempting to produce the final answer in a single step. This decomposition allows the model to consider intermediate facts and temporal constraints explicitly, which improves the accuracy of the final answers. In contrast, GPT-4o tends to generate a list of responses that are partially correct, incomplete, or even irrelevant. Furthermore, after fine-tuning, PoK's outputs align more closely with standard answer formats. GPT-4o, however, often produces semantically correct answers that fail to match the required format, which leads to incorrect evaluation scores. For example, when the ground-truth answer is "2012-05" for a question such as "in which month...", GPT-4o frequently

TABLE XI
EXAMPLE OF A 2-HOP REASONING PROCESS IN PoK AND GPT-4o

Type	After & first, 2-hops
Question	After Okada Katsuya, who wish to visit Cambodia first?
Answer	Foreign Affairs (South Korea)
Plan	(1) Retrieve & Reason: [time] = When Okada Katsuya visits Cambodia? (2) Retrieve: Who wishes to visit Cambodia first after [time]? (3) Rank by timestamps in ascending order. (4) Reason: Who wishes to visit Cambodia first after [time]?
PoK Response	(1) ['2009-10-02'] (2) ['Foreign Affairs (South Korea)']
GPT-4o Response	- South Korea - Thailand -Foreign Affairs (South Korea)

TABLE XII
EXAMPLE OF THE ERROR IN PoK

Question	After the Cabinet Council of Ministers of Kazakhstan, who was the first to express the intention to negotiate with China?
Answer	Foreign Affairs (South Korea)
Plan	(1) Retrieve & Reason: [time] = When did the Cabinet Council of Ministers of Kazakhstan happen? (2) Retrieve: Who expresses the intention to negotiate with China after [time]? (3) Rank by timestamps in ascending order. (4) Reason: Who is the first?
PoK Response	(1) 2007-04-24 (2) Japan

outputs "May," which, although semantically valid, is considered format-inconsistent and thus counted as incorrect during evaluation.

2) *Error Analysis*: To better understand the limitations of PoK, we conduct a comprehensive error analysis from both quantitative and qualitative perspectives. As shown in Table XIII, the correctness of PoK's final answers is highly dependent on retrieval quality. Although there are occasional cases where the model produces correct answers without successfully retrieving the relevant knowledge, this is mainly attributable to the implicit knowledge stored within LLMs, rather than robust reasoning over external evidence. This further highlights the model's reliance on parametric knowledge when

TABLE XIII

ERROR STATISTICS (%). WE CATEGORIZE THE OUTCOMES INTO FOUR STATES BASED ON WHETHER THE SUPPORTING EVIDENCE IS *RETRIEVED* OR *UNRETRIEVED*, AND WHETHER THE PREDICTION IS *CORRECT* OR *INCORRECT*

Dataset	Retrieved		Unretrieved	
	Correct	Incorrect	Correct	Incorrect
MULTITQ	76.5	10.0	1.4	12.1
TimeQuestions	67.0	9.1	16.2	7.7
Timeline-ICEWS	33.2	9.9	27.0	29.9
Timeline-CronQuestion	36.4	11.7	28.7	23.2

retrieval fails. Table XII presents a representative failure case. In the first step, the model generates an incorrect sub-question due to hallucination, omitting critical information. Consequently, it fails to retrieve the correct timestamp and instead retrieves incorrect facts. This error then propagates through the subsequent *Retrieve*, *Rank*, and *Reason* stages, ultimately leading to an incorrect final answer. This case illustrates how early-stage errors can cascade through the pipeline, emphasizing the importance of reliable decomposition and retrieval for ensuring overall reasoning accuracy.

VI. CONCLUSION AND FUTURE WORK

In this work, we address two core challenges faced by LLMs in handling temporal questions and propose PoK, a Time-aware Plan of Knowledge framework. By decomposing complex temporal questions into structured sub-objectives grounded in pre-defined operators (*Retrieve*, *Rank*, and *Reason*), PoK explicitly decouples temporal ordering from semantic inference at the reasoning level, mitigating hallucination in multi-hop reasoning. Complementarily, a time-aware retrieval module ensures that each retrieval step remains sensitive to temporal constraints at the representation level, supplying the reasoning process with semantically and temporally aligned evidence via a fine-tuned Temporal Knowledge Store (TKS). Extensive experiments on four benchmark TKGQA datasets verify the effectiveness of PoK, achieving absolute performance gains of up to 35.3 percentage points.

Despite these improvements, reasoning over incomplete TKGs remains a significant challenge, where missing temporal facts hinder reliable multi-hop inference. Future work should focus on developing more robust mechanisms for handling incomplete temporal knowledge and improving the retrieval of implicit or missing temporal evidence. In addition, errors in question planning may propagate through subsequent retrieval and reasoning stages, leading to incorrect final predictions. Future work can therefore further investigate more reliable planning strategies, such as incorporating self-verification to reduce cascading errors in multi-step temporal reasoning.

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Ying Zhang received the PhD degree in computer science from Nankai University, China, in 2013. From 2011 to 2013, she studied with Purdue University, West Lafayette, IN, USA. She is currently a professor with the College of Computer Science, Nankai University. Her main research interests include sentiment analysis, multimodal data analysis, and information retrieval.



Xinying Qian received the BM degree in information management and information system from the Zhongnan University of Economics and Law, China, in 2023. She is currently working toward the PhD degree with the College of Computer Science, Nankai University, China. Her current research interests include knowledge graph reasoning, and natural language processing.



Yu Zhao received the BE degree in software engineering from Nankai University, China, in 2017. She is currently working toward the PhD degree with the College of Computer Science, Nankai University, China. Her current research interests include multimodal knowledge graphs, knowledge representation and reasoning, and natural language processing.



Baohang Zhou received the PhD degree in computer science from Nankai University, China, in 2025. From 2023 to 2024, he studied with the Chinese University of Hong Kong, China. He is currently a lecturer with the School of Software, Tiangong University. His main research interests include multimodal knowledge graphs, data mining, and large language models.



Xiaojie Yuan received the PhD degree in computer science from Nankai University, China, in 2000. She is currently a professor with the College of Computer Science, Nankai University. She leads a research group working on topics of databases, data mining, and information retrieval.



Xuhui Sui received the PhD degree in computer science from Nankai University, China, in 2026. His current research interests include entity linking, knowledge graphs, and information extraction.